

JEE MAIN REVISION MASTER: BASICS OF ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 295 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Basics Of Organic Chemistry - Core Application	295 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basics Of Organic Chemistry - Core Application (Very High Yield — 295 PYQs)

Core Principle: Key problem pattern in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Statement I : Vapours of the liquid with higher boiling point condense before vapours of the liquid with lo..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Reaction List-I Mechanism A. Williamson Synthesis I. Electrophilic addition B. Friedel Craft R..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The strongest conjugate acid will result from :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 23rd January Morning Shift (Q#1)

Given below are two statements : Statement I : Sublimation is used for the separation and purification of compounds with low melting point. Statement II : The boiling point of a liquid increases as the external pressure is reduced. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are false **[Correct]**
- (B) Statement I is true but Statement II is false
- (C) Both Statement I and Statement II are true
- (D) Statement I is false but Statement II is true

Correct Answer: A

Solution Analysis: Statement I: False. Sublimation is used to separate/purify sublimable (volatile) solids —solids that pass directly from solid to vapour on heating. This depends on having sufficient vapour pressure in the solid state, not simply on having a low melting point . Statement II: False. The boiling point decreases when external pressure is reduced (principle of vacuum distillation). ■ Correct option: A (Both Statement I and Statement II are false).